

HOOPER BAY, AK, USA

WMO: 702186

Lat: 61.524N

Lon: 166.147W

Elev: 18

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26651

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.5	-10.8	-26.8	1.2	-10.2	-23.5	1.5	-5.0	43.0	22.8	39.7	20.2	10.6	110	0.970

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.4	65.7	57.1	62.8	55.1	59.0	54.2	58.6	63.7	56.7	61.1	54.8	58.3	11.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.5	65.5	59.8	54.4	62.9	58.8	52.7	59.0	57.2	25.4	63.7	24.1	61.0	23.2	58.3	69.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
35.9	32.5	28.7	DB	-18.1	70.0	7.7	5.0	-23.7	73.5	-28.2	76.4	-32.5	79.2	-38.1	82.8	
			WB	-19.1	60.5	6.1	4.8	-23.6	63.9	-27.1	66.7	-30.6	69.4	-35.0	72.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.1	11.8	15.4	13.4	25.9	36.2	46.3	51.4	51.9	46.6	34.7	24.2	14.2
	DBStd	17.65	14.72	12.42	11.97	10.18	7.33	5.87	4.55	4.63	5.60	7.87	10.26	11.99
	HDD50	7138	1187	969	1134	725	431	140	36	28	128	476	775	1109
	HDD65	12387	1650	1389	1599	1172	893	563	423	407	553	939	1225	1574
	CDD50	230	3	0	0	3	4	28	78	87	25	2	0	0
	CDD65	1	0	0	0	0	0	0	0	1	0	0	0	0
	CDH74	5	0	0	0	0	0	2	1	2	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	14.8	16.3	17.8	15.9	14.4	12.9	12.9	12.4	13.7	14.4	14.2	16.0	16.7
Precipitation	PrecAvg	28.1	1.9	1.9	1.3	1.2	1.5	2.1	3.0	3.9	3.8	2.8	2.6	2.1
	PrecMax	38.3	4.3	4.6	4.2	2.4	3.2	3.2	4.6	5.8	6.7	5.6	5.1	4.4
	PrecMin	22.6	0.4	0.6	0.2	0.3	0.2	0.9	1.8	1.3	0.6	0.8	0.5	0.5
	PrecStd	3.7	0.9	1.0	0.9	0.5	0.7	0.6	0.8	1.0	1.1	1.0	1.2	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.8	39.4	44.6	57.4	61.3	69.5	69.9	71.8	65.7	56.8	50.1	42.6
		MCWB	41.0	35.1	35.9	45.5	49.7	56.6	61.0	61.5	55.8	45.4	40.7	36.8
	2%	DB	45.6	35.5	35.4	51.5	53.7	61.0	63.8	65.8	63.2	51.9	46.0	35.1
		MCWB	37.6	33.5	32.3	41.8	47.0	53.7	57.0	58.3	53.4	43.3	38.2	33.3
	5%	DB	36.1	33.7	33.2	38.9	49.1	56.7	60.9	61.3	56.5	48.0	39.2	33.6
		MCWB	33.4	32.2	31.5	35.7	44.9	51.5	56.3	57.1	52.4	44.0	36.5	31.9
	10%	DB	31.7	31.6	30.4	36.2	45.7	54.2	57.4	57.5	53.9	45.4	36.6	30.4
		MCWB	30.6	30.8	29.3	33.9	42.3	50.4	53.7	54.2	50.9	42.5	34.9	29.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.1	35.3	36.2	46.4	51.5	57.6	62.2	62.1	56.7	48.2	42.2	37.1
		MCDB	58.4	39.0	42.0	58.7	59.1	65.8	67.2	68.4	62.7	51.4	46.5	42.6
	2%	WB	37.2	33.1	32.9	41.6	48.0	54.5	58.5	59.1	54.2	46.2	39.0	33.3
		MCDB	46.0	34.2	34.4	50.7	52.9	59.3	62.9	64.1	60.0	49.1	43.8	34.6
	5%	WB	33.3	32.0	31.6	36.3	45.3	52.2	56.4	57.3	52.6	44.3	37.0	31.9
		MCDB	35.7	33.1	32.6	39.0	48.9	56.2	60.5	61.8	57.1	47.2	39.7	32.9
	10%	WB	30.4	30.3	29.5	33.6	42.4	50.3	54.5	55.2	50.6	42.3	34.8	30.1
		MCDB	31.5	31.2	30.5	35.3	45.8	54.0	57.6	58.1	53.5	46.0	37.0	30.6
Mean Daily Temperature Range	5% DB	MDBR	10.8	9.8	11.8	10.0	7.7	8.2	6.8	6.4	6.9	6.9	8.2	9.6
		MCDBR	16.0	9.6	12.6	11.3	14.3	14.5	12.3	11.7	10.0	6.3	9.0	8.7
		MCWBR	12.5	8.8	10.8	8.1	9.7	9.2	7.8	7.5	6.1	5.1	7.8	8.1
	5% WB	MCDBR	14.4	9.3	11.0	10.7	13.5	13.3	11.0	10.6	8.0	5.6	7.8	8.7
MCWBR		11.8	9.1	9.8	7.7	9.5	8.8	7.5	7.0	5.6	5.3	7.1	8.4	
Clear-Sky Solar Irradiance	taub	0.222	0.285	0.290	0.355	0.373	0.374	0.381	0.395	0.345	0.309	0.264	N/A	
	taud	2.037	2.110	2.185	2.045	2.212	2.395	2.416	2.336	2.453	2.335	2.036	N/A	
	Ebn at Noon	155	213	261	261	267	271	265	249	244	207	127	N/A	
	Edh at Noon	12	23	30	43	39	34	32	32	24	18	12	N/A	
All-Sky Solar Radiation	RadAvg	100	336	882	1413	1585	1630	1329	1046	678	331	104	44	
	RadStd	15	43	105	134	199	144	99	136	60	25	16	6	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page